Mission Description Document

SQRT (Sub-zero Quantitative Research Team)

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Key Reference Documents

1. SQT-PDR-PMP-0
2. SQT-PDR-SEP-0
3. SQT-PDR-DJF-0
4. UCD-DRD-2026-0
5. UCD-CFP-2026-0

1 Executive Summary

This Mission Description Document presents the SQRT mission, including our Mission Context, Mission Goals and Mission Concept.

Mission Statement: The SQRT mission will produce and observe the flash evaporation of water via exposure to a low pressure and low temperature environment to answer the question of how liquid plumes on icy moons thermally evolve.

The mission is designed to investigate the thermal evolution of water when exposed to extremely reduced-pressure conditions during flight at altitudes between 20km and 25km. The mission focuses on evaporation-driven cooling and plume-like vapour behaviour that arise as pressure decreases away from sea-level conditions. Rather than attempting to directly recreate cryovolcanic plumes, the experiment isolates the fundamental physical response of a water sample to a rapid pressure drop in a controlled and interpretable way.

The scientific motivation for this study is informed by plume observations on icy moons such as Enceladus [3, 4, 6, 7] and Europa [1, 8], where pressure-sensitive fluids are believed to form vapour structures when exposed to low-pressure environments. While the geological and subsurface mechanisms responsible for these plumes are highly complex, a common underlying element is the thermal evolution of material undergoing rapid, or flash, evaporation. The phase diagram of water shown in Figure 1 highlights how strongly its physical state depends on temperature and pressure, providing a clear physical basis for studying these processes in a reduced-pressure environment.

Based on this context, the primary objective of the SQRT mission is to investigate the thermal evolution of water when exposed to reduced pressure under realistic, time-varying flight conditions. A secondary objective is to observe evaporation-driven cooling and plume-like behaviour in a non-laboratory environment, while demonstrating that such processes can be studied using a simple, contained experiment compatible with a TupperSat platform. These objectives inform the definition of a set of Level 1 mission requirements that specify the measurable performance needed to achieve mission success, while remaining independent of specific implementation choices.

The mission concept describes how these objectives and requirements are achieved during flight. As the balloon ascends to stratospheric altitudes, ambient pressure and temperature decrease continuously, providing a naturally evolving low-pressure environment. The TupperSat payload contains a water sample housed within an insulated container and connected via a valve to an adjoining low-pressure observation chamber that equilibrates with the external atmosphere. At a predetermined trigger condition, such as altitude or ambient pressure, the water is exposed to the reduced-pressure chamber, initiating rapid evaporation and associated cooling. This transient, plume-like behaviour is monitored using onboard temperature and pressure measurements, with optical imaging used where feasible to support visual interpretation.

Data collected during flight are compared with ground-based reference tests conducted at sea-level pressure to isolate the effects of pressure reduction on thermal behaviour. Throughout the mission design, priority is given to controlled activation, clear measurement, and straightforward analysis, while remaining within the mass, power, volume, and operational constraints of the TupperSat platform. Overall, the SQRT mission demonstrates a simple, robust, and feasible approach to studying evaporation-driven thermal evolution in low-pressure environments using a high-altitude balloon system.

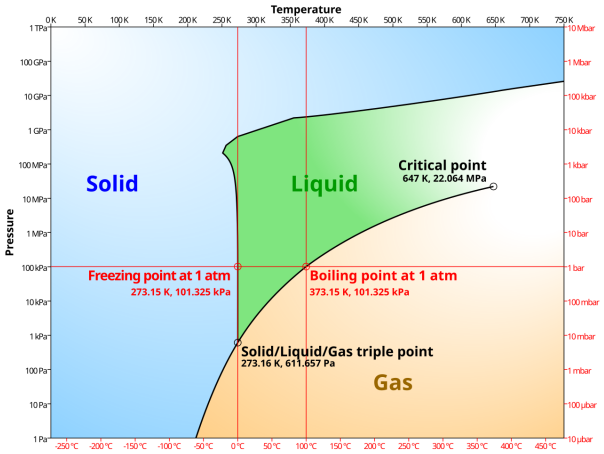


Figure 1: Phase Diagram of Water [2]

2 Mission Context

The SQRT mission is inspired by the plumes observed on icy moons such as Enceladus [7, 4, 3, 6] and, more recently, Europa [1, 8]. These plumes are thought to have formed when internal fluids are suddenly exposed to very low-pressure environments, which then causes a rapid evaporation and cooling as the system moves away from equilibrium [10, 9]. While the full processes responsible for these plumes involve complex geological and subsurface mechanisms, a key part of how they occur is the thermal response and phase change of water undergoing a rapid pressure drop [5].

Rather than attempting to replicate plumes directly, this project focuses more on isolating and studying this initial phase change. The main objective is to observe the thermal evolution of a sample of water as it is suddenly exposed to a reduced pressure and temperature environment, and to investigate how evaporation-driven cooling and vapour formation develop over time. By concentrating on evaporation and phase changes, the experiment avoids over-complicating the system while still capturing plume-like characteristics/analogues in a controlled and interpretable way. This high-altitude balloon TupperSat flight provides us with a practical and realistic environment for this type of investigation. At altitudes of 20–25 km, the pressure is much lower than at sea level (50 mbar and 25 mbar, respectively), resulting in much lower boiling points for water (32.9°C and 21.1°C, respectively). The external environment is also far colder, approximately -40°C to -60°C, and experiences much higher radiation levels than those here at ground level. The payload will operate in a non-sterile, "real-world" flight environment, where temperature, pressure, and motion change continuously and unpredictably during ascent. This makes the experiment an interesting alternative to laboratory setups for studying how water responds thermally under dynamic, low-pressure conditions.

During the early stages of mission planning, more direct ejection-based plume concepts were explored. Although after discussing this idea with Dr. Hans Huybrighs (Dublin Institute for Advanced Studies) `https://hanshuybrighs.com/`, the mission was refocused toward a contained experiment that exposes water to reduced pressure within a controlled chamber. This approach was considered more viable as it's more robust, physically interpretable, and measurable, while still preserving the plume-like aspect of rapid vapour formation. It also reduces risk and increases the likelihood of acquiring usable data within the unpredictable and inertial motion constraints of a balloon and wind motion.

The SQRT mission aims to track how a small sample of water evolves thermally as pressure decreases during altitude ascent. By recording temperature, pressure, and altitude results, the experiment will investigate how evaporation cooling progresses within a low-pressure environment. Ground-based tests conducted at sea-level pressure rates will be used as a reference, allowing the effects of reduced pres-

sure to be isolated from other environmental influences. This mission is motivated by achieving clear measurements, controlled activation, and straightforward analysis. The results shall demonstrate how plume-like thermal evolution and evaporation behaviour can be studied in low-pressure, high-altitude environments using a high-altitude balloon with an attached TupperSat.

3 Mission Goals

- The primary objective of the SQRT mission is to investigate how a water sample physically reacts when it has been exposed to a reduced-pressure environment during a high-altitude balloon ascent. In particular, the mission aims to examine evaporation-driven cooling and plume-like vapour behaviour that occur as pressure decreases away from sea-level conditions.
- A secondary goal of the mission is to observe how this thermal evolution develops over time in a dynamic flight environment, where pressure, temperature, and altitude are continuously changing. By recording time and altitude dependent behaviour, our mission hopes to investigate how rapidly evaporation and associated cooling effects occur under realistic high-altitude conditions.

Together, these combined goals aim to create a clear link between instant pressure drop exposure event and a the expected plume-like reaction of sample of water, while remaining feasible and cost-efficient within the mass, power, volume, and operational constraints of the TupperSat system.

4 Mission Requirements

The following mission requirements found in Table 1 define the specific, measurable performance criteria that the TupperSat payload shall satisfy in order to achieve the SQRT mission goals.

Table 1: SQRT Level 1 Mission Requirements

Requirement ID	Definition	Value	Units	Justification
SQT:MR:010	The TupperSat shall measure the temperature of the water and/or surrounding vapour during exposure to reduced pressure.	≥ 0.1	°C	To characterise and quantify its thermal evolution.
SQT:MR:020	The TupperSat shall measure pressure within the low-pressure chamber throughout the flight.	≥ 5	mbar	Allow correlation between chamber pressure and thermal behaviour across ascent.
SQT:MR:030	The TupperSat shall time-tag all primary measurements.	20	s	To correlate thermal evolution as a function of time.
SQT:MR:040	The TupperSat shall record altitude data with all primary measurements.	50	m	To correlate thermal behaviour with external conditions during ascent.
SQT:MR:050	The TupperSat shall provide verifiable confirmation that the exposure event occurred during flight.	N/A	N/A	In case of non-recovery, confirms experimental execution.

SQT:MR:060	The TupperSat shall maintain the water sample within the low-pressure chamber for a time sufficient to observe thermal evolution after exposure.	N/A	N/A	Ensures the acquisition of sufficient data for thermal evolution studies. The specific time can be defined once the camera is procured and the frame rate is a known variable.
SQT:MR:070	The TupperSat shall record continuous external pressure data during the exposure period.	≥ 5	mbar	To aid post-mission analysis and compare external pressures to those inside the low-pressure chamber to appraise vent efficiency.
SQT:MR:080	The TupperSat shall be designed in such a way that primary experimental data is recorded, stored and retrievable following payload recovery.	N/A	N/A	To complete post-recovery analysis.
SQT:MR:090	The TupperSat shall be designed to allow repeatable ground-based testing.	N/A	N/A	To ensure reliability and durability of mission-critical subsystems.
SQT:MR:100	The TupperSat shall be designed in such a way to expose water at altitudes between 20-25 km in altitude.	N/A	N/A	To ensure the water is exposed at appropriate external conditions and that flash-evaporation will occur.

5 Mission Concept

The SQRT mission is based on a simple operational principle: the equilibrium state of a small sample of liquid water will change when exposed to a reduced-pressure environment, leading to its rapid evaporation. After some time the evaporated water will then cool. By controlling when and where this pressure exposure occurs during high-altitude ascent, the experiment hopes to directly observe how water thermally evolves as it transitions away from sea-level equilibrium conditions. The mission is flown on a high-altitude balloon using the TupperSat platform, which naturally provides a continuously decreasing pressure environment during ascent. As the TupperSat rises to stratospheric altitudes of approximately 20–35 km, its surrounding pressure drops significantly which reduces the boiling point of the water. This pressure reduction, instead of the external heating, is the main cause of evaporation and its plume-like vapour creation in the experiment.

To use this environment in a properly controlled way, the payload contains a water sample held in an insulated cylindrical container and is connected using a controlled valve to an observation chamber that is pressure equivalent to the external atmosphere. At a predefined trigger point condition, the water is exposed to the low-pressure environment within the observation chamber. We expect that this sudden change in pressure would almost instantly cause a rapid evaporation. The thermal evolution during this evaporation flash and subsequent cooling will be recorded with a camera capturing IR wavelengths. If the set-up conditions are suitable, a second camera may be set up to capture the evolution from two points of view. optical. If the set-up conditions are suitable, the mission may also record visible vapour or condensation. Figure 2 on the next page shows an initial conceptual sketch for the SQRT TupperSat, including a 3D based model outlining the key relevant apparatus required by the SQRT payload: the segregated payload chambers. The low pressure chamber is exposed to the pressure of the external environment, and the other section of the TupperSat contains the sealed water container (connected to the low pressure chamber via valve) and the electronics bus. A more detailed initial design of the TupperSat is included in SQT-PDR-DJF-0.

The core mission goal is simply characterising the thermal evolution of water under reduced pressure. We plan to achieve this by continuously monitoring temperature and pressure before, during, and after the actual exposure and release moment. Time and altitude based measurements allow the thermal response to be correlated directly with changing external pressure, which shall allow the isolation of pressure-driven effects from other environmental factors. Optical observations would provide supporting evidence of plume-like behaviour, but are secondary goals to the thermal and pressure measurements. The mission will directly measure how an instant reduction in pressure alters the temperature and phase of water. In this way, the operational principle of controlled pressure exposure is closely linked to the mission goals: the balloon ascent provides the environment, the exposure event initiates the physical process of interest, and the measurements capture the resulting thermal evolution.

The mission concept prioritises simplicity, containment, and clear physical interpretation. This method of experimental approach ensures that the experiment remains feasible and realistic within the limits of the TupperSat platform while producing data that directly supports the mission goals.

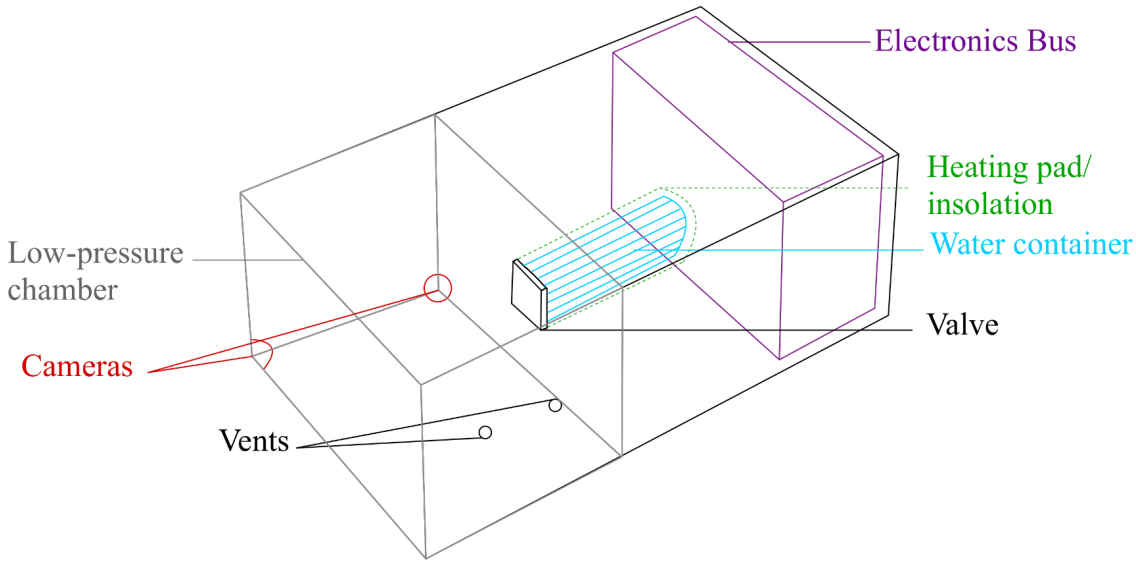


Figure 2: SQRT TupperSat Conceptual Sketch

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